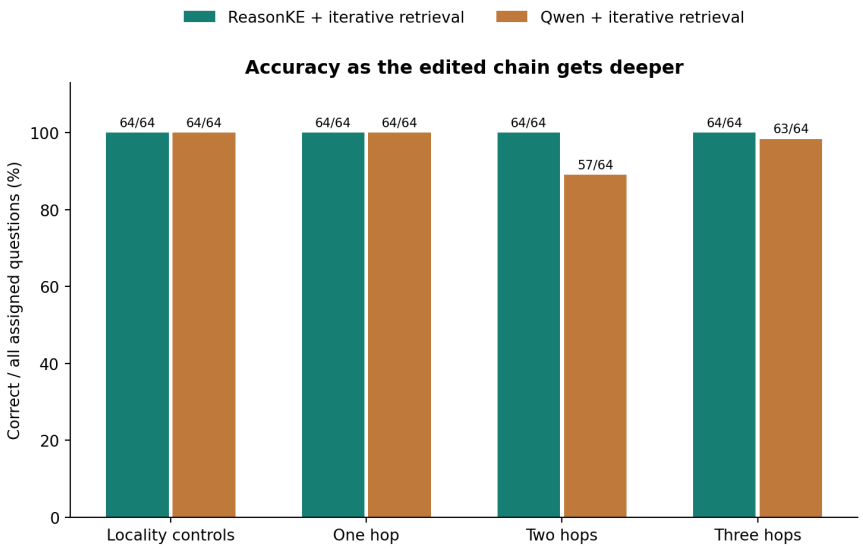
Can an edit travel through three facts?

A measured addendum to the completed operations study. An ownership correction must follow service -> team -> on-call person -> office city, while historical-release and other-property answers remain correct.

Final requests	Main questions per model	Oracle per model	Lineage units
544	256	16	32

Configuration	Correct / assigned	Wrong	Abstain	Errors	Unexecuted
ReasonKE + retrieval	256/256	0	0	0	0
Qwen + retrieval	248/256	0	8	0	0
ReasonKE + gold support	16/16	0	0	0	0
Qwen + gold support	16/16	0	0	0	0

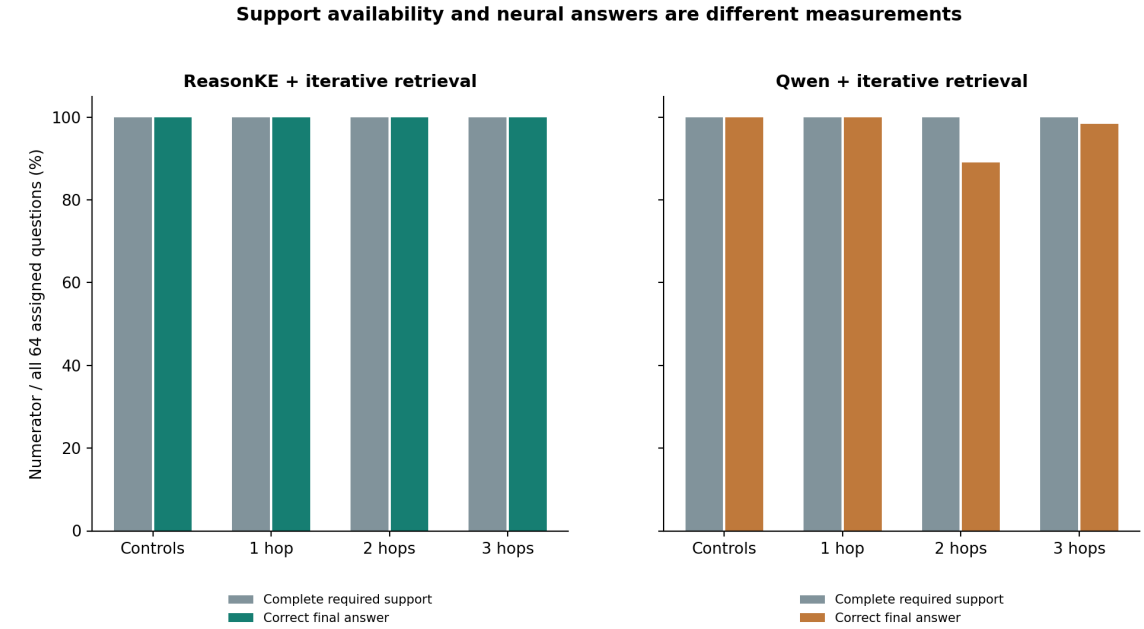


64 questions per depth and model; locality controls are separate-property/version questions. All outcomes retain denominator credit.

Primary accuracy is correct / all assigned. Abstentions, errors and unexecuted assignments receive zero primary credit. These are 256 correlated question variants over 32 graph lineages, not 256 independent graphs. The original operations-v1 results stay unchanged and are not pooled with this study.

Separate support from reasoning

The same bounded lexical selector serves both checkpoints: four facts, two question-based seeds and at most three answer-frontier expansion rounds. It reads approved text rather than expected answers or gold support IDs. A development-only comparison selected this policy before final inference.



Both models use the same selector. Repeated use is not two independent retrieval experiments; missing/unexecuted retrieval stays in the denominator.

Checkpoint / three-hop main	Required facts found	Complete paths	Correct answers
ReasonKE + retrieval	192/192	64/64	64/64
Qwen + retrieval	192/192	64/64	63/64

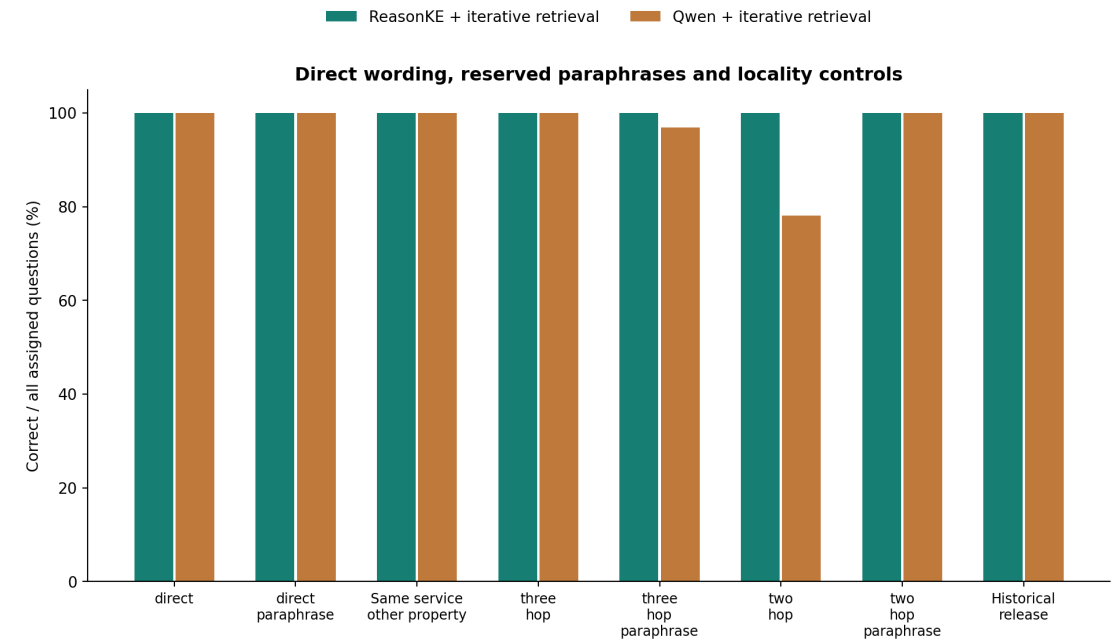
Gold-support diagnostics

Sixteen three-hop questions per model were selected by a fixed hash before inference. Their gold facts isolate answer generation from ordinary retrieval. This subset is not targeted failure recovery and does not replace the 256-question primary denominator.

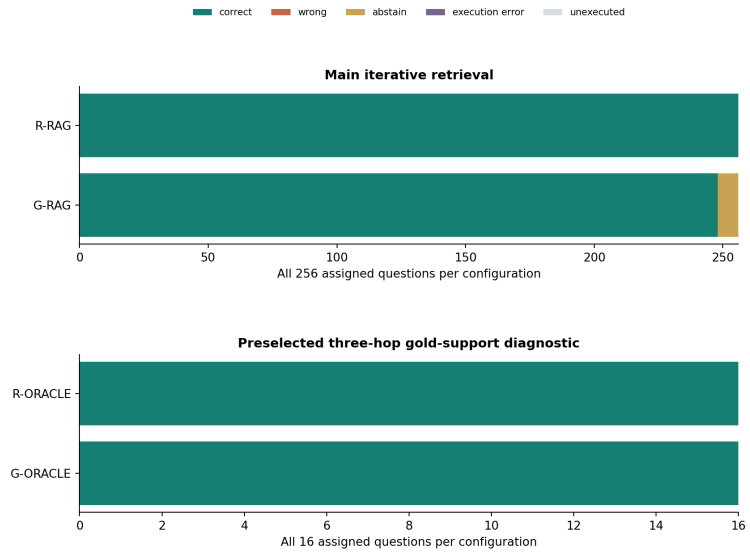
ReasonKE minus Qwen	Observed difference	95% paired interval	Lineages
RAG	+3.12 pp	[+1.56, +5.08] pp	32
RAG-depth-3	+1.56 pp	[+0.00, +4.69] pp	32
ORACLE	+0.00 pp	[+0.00, +0.00] pp	13

Paired bootstrap resamples whole lineages (10,000 draws, seed 42). A zero-width interval only describes the sampled lineage differences; it does not establish certainty for arbitrary questions. Reusing one selector with two models is not two independent retrieval experiments.

Wording, scope and every outcome



32 assigned questions per category and model. Wording variants remain correlated within each of 32 held-out lineages.

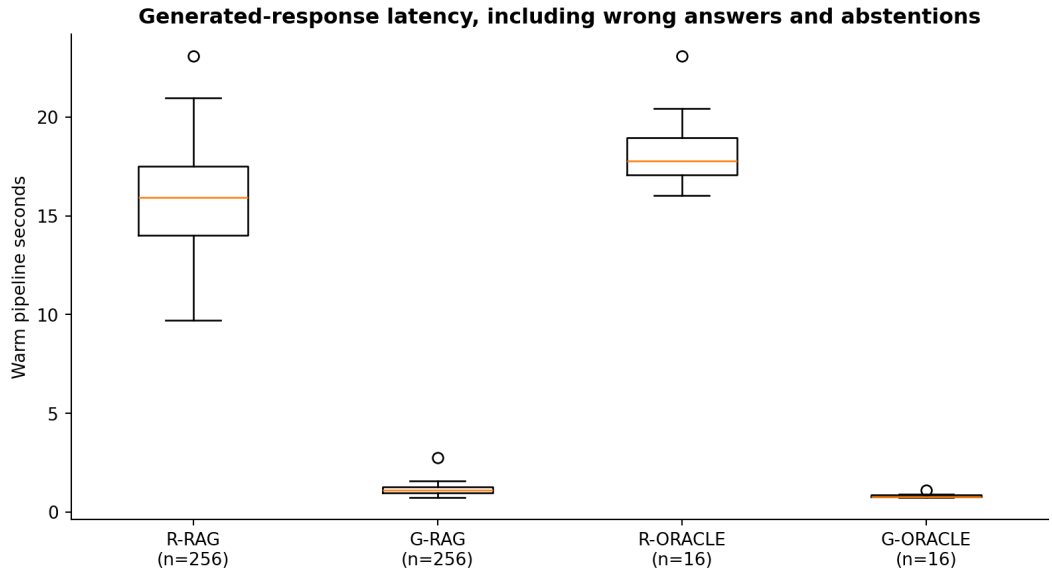


The packet contains 48 fictional Indian-context lineages and twelve facts per world. Sixteen development lineages and 32 final lineages were split before question augmentation. Wording-template banks, service/team/person identities and aliases are separated by split; city labels and relation vocabulary repeat. The service and workplace assignments are invented.

Each final lineage supplies eight questions: direct and paraphrased questions at depths one, two and three, plus two locality controls. Depth zero labels controls, not a zero-fact task: each still needs one separate property or historical fact. There are 64 assigned questions per depth group and 32 per wording category, per main condition.

Scoring checks the whole endorsed final answer against controlled aliases. Mentioning the target in a rejected option is insufficient. Strict format scoring can reject an otherwise informative response. A correct final answer does not certify the model's generated explanation. No sole LLM judge, independent human semantic-review claim or arbitrary-language verification is used.

Measured cost and finite execution



Selection + local IPC + generation; excludes model loading, downloads and browser network. Errors/unexecuted disclosed separately.

Configuration	Generated n	Warm p50 / p95 seconds	Coverage
ReasonKE + retrieval	256	15.94 / 19.28	100.0%
Qwen + retrieval	256	1.12 / 1.38	96.9%
ReasonKE + gold support	16	17.77 / 21.12	100.0%
Qwen + gold support	16	0.76 / 0.95	100.0%

Warm pipeline latency includes context selection, local IPC and generation, including generated wrong answers and abstentions. It excludes loading, downloads, browser network and administrator actions. Error latency has its own saved population.

Checkpoint	Peak loading allocation GiB	Peak generation allocation GiB
ReasonKE	6.59	5.39
Qwen	6.59	5.39

Both pinned 7.62B-parameter checkpoints use NF4 double quantization, FP16 computation, greedy decoding, a 4,096-token input limit and a 512-new-token cap. They were trained upstream. This project performed no checkpoint fine-tuning or per-correction weight update. The instruction checkpoint is a matched comparison, not a verified causal ablation of ReasonKE training.

Profiles: eight requests per checkpoint. Development: 24 per checkpoint. Final: 272 per checkpoint. Every outcome is written atomically; exact-plan resume skips recorded outcomes. An earlier development dispatch lost its temporary Colab runtime before export; its unknown outcomes are excluded from measured evidence. Preserved profiles were restored unchanged before a disclosed replacement development run.

Conclusion and evidence trail

On the frozen depth population, ReasonKE with iterative retrieval scored **256/256**; Qwen with the same retrieval scored **248/256**. These measured outcomes and their latency/support trade-offs are the conclusion of this experiment. No universal edit success, arbitrary paraphrase robustness, faithful reasoning trace or superiority outside this controlled population is inferred.

Observed non-correct outputs

Record	Outcome	Expected	Final answer / error
G-RAG:depth-001-two_hop	abstain	Diya Desai	UNKNOWN
G-RAG:depth-004-two_hop	abstain	Arjun Desai	UNKNOWN
G-RAG:depth-016-two_hop	abstain	Arjun Nair	UNKNOWN

Up to three lexicographically first non-correct records are shown, without selection for narrative effect. The raw ledgers preserve every output and error.

Reproduce and inspect

Artifact / command	What it establishes
protocol.json + six raw ledgers	Frozen source/data/configuration identities; original profile, development and all assigned final outputs
host-audit.json + derived/figure-manifest.json	Byte-bound mechanical audit and scientific export provenance; zero new model calls
scorecard.csv / retrieval.csv / latency.csv	Outcome denominators, required support, latency samples, tokens and allocated/reserved memory
python -m scripts.build_depth_report	Audits final receipts before regenerating tables and five figure pairs
docs/UNDERSTAND_DEPTH_EXTENSION.md	Plain-language history, interpretation and India demonstration workflow

Sources and attribution

YukinoKaren/ReasonKE_plus
Pinned revision: ceaedc2ae7c8fed6257953587d991ce550db6106

Qwen/Qwen2.5-7B-Instruct
Pinned revision: a09a35458c702b33eeacc393d103063234e8bc28

EditGuard repository | Reason-KE official code | Reason-KE++ paper. Published scores are not substituted for these measurements. The released ReasonKE weight-license metadata was not established; weights are fetched from their pinned upstream source rather than redistributed in the website container.

Generated UTC: 2026-10-28T15:53:51.143262+00:00
Results SHA-256: 3501e711f6213b9984b115d354c87ed6b2ab6c2b50d53a85f693b9e7f2dc7db7